MIAMI UNIVERSITY | COLLEGE OF ENGINEERING AND COMPUTING
Department of Computer Science and Software Engineering

CSE 434 / CSE 534: Generative Artificial Intelligence

Spring 2026 (Hybrid Synchronous)

Instructor: Dr. Yang Zhang
Email: Zhang981@miamioh.edu
Website: https://yznd42.github.io/
Term: Fall 2025
Meeting Times:
Session A: Mondays and Wednesdays, 2:50 PM - 04:10 PM
Session B: Mondays and Wednesdays, 1:15 PM - 2:35 PM
In Person Meeting (Jan. 26 to Feb. 8, 2026, and May 2 to 15, 2026): 166 McVey Data Science Building
Online Zoom Meeting (Feb. 9 to May 1, 2026):
https://miamioh.zoom.us/j/87373270550?pwd=zPEzgbIICWsKzYAT3eYvta80WHfTs7.1

Course Overview

This course introduces the fundamental concepts and techniques of Generative AI, with a focus on language models, neural networks, and creative AI systems. Students will explore state-of-the-art generative models such as GANs, transformers, diffusion models, and VAEs, and learn how to apply these techniques to real-world problems. The course progresses from hands-on exploration of modern tools to deep mathematical foundations, culminating in advanced research topics and independent projects. Emphasis will be placed on prompt engineering, model fine-tuning, evaluation, and responsible AI practices.

Prerequisites:
Familiarity with Python programming is required. Students are encouraged (though not required) to have completed an introductory course in machine learning or AI.

Course Format:
Lectures will be complemented by hands-on assignments, quizzes, discussions, and a course project. The course does not include a midterm or final exam. Graduate students (CSE 534) will have additional requirements, including advanced readings, leading discussions, deeper engagement with generative model frameworks, and a more substantial course project.

Contact Information

Instructor: Dr. Yang Zhang
Email: zhang981@miamioh.edu
Office Hours: Available on Zoom/in person by appointment via email

TA: Manish Bhandari
Email: bhandam4@miamioh.edu
Office Hours: TBD after student survey

Course Communication: All important announcements, assignments, and updates will be posted on Canvas. Please check Canvas regularly and enable notifications.

Student Learning Outcomes (SLOs)

By the end of this course, students will be able to:

1. **LO1:** Describe how generative AI relates to other areas of machine learning and can be used to perform specific tasks.
2. **LO2:** Implement Python code to generate data using modern programming environments (e.g., Jupyter Notebooks, APIs from modern libraries) to generate text, images, and/or numerical data. Describe key concepts in data generation, including oversampling, fairness and bias, conditioning, alignment, and hallucination.
3. **LO3:** Evaluate the quality of generated AI solutions using quantitative and qualitative metrics. Implement and apply quantitative algorithms to evaluate the quality of generated data, such as overlap, semantic similarity, or n-gram matches. Apply qualitative assessment processes (e.g., for fluency, faithfulness, coverage) and determine the inter-rater agreement.
4. **LO4:** Optimize existing AI solutions to deliver expected results using prompt engineering and fine-tuning to identify the best generator's hyper-parameters to generate an AI output that performs best on an application scenario (e.g., question-and-answering system). Extend a knowledge model with external data.

Graduate Section (CSE 534 A) Additional Expectations:

- **LO5 [Graduate]:** Adapt methods and metrics from recent research articles (beyond what is covered in class) to develop robust, fine-tuned, generative AI models.
- **LO6 [Graduate]:** Demonstrate an ability to do independent research in identifying and applying state-of-the-art metrics to assess and evaluate generative AI models.

Materials Needed

Recommended (Not Required) Texts:

- Raschka, Sebastian. *Build a large language model (from scratch).* Simon and Schuster, 2024.
- Foster, David. *Generative deep learning.* O'Reilly Media, Inc., 2022.
- Wang, Dong, Lanyu Shang, and Yang Zhang. *Social Intelligence: The New Frontier of Integrating Human Intelligence and Artificial Intelligence in Social Space.* Springer Nature, 2025.

Supplementary Resources:
For each class, related readings (including research papers, tutorials, and other resources) will be provided in advance on Canvas.

Software:
Python, TensorFlow/PyTorch, OpenAI, Gemini API etc. (setup instructions will be provided on Canvas).

Grading Scale

Your final letter-grade in this course will be determined as shown in the table below:

97–100 = A+
94–96.99 = A
90–93.99 = A-
87–89.99 = B+
83–86.99 = B
80–82.99 = B-
77–79.999 = C+
73–76.99 = C
70–72.99 = C-
67–69.999 = D+
63–66.99 = D
60–62.99 = D-
59.99 and below = F

- All course grades will be posted in the Canvas. Instructor reserves the right to curve grades at his discretion to help the students, based on class performance. Instructor will try to ensure that the grade earned by a student never reduces as a consequence of the curving.
- Grade adjustments will be limited to automatic rounding. Fractional values will be rounded automatically. Fractional portions of grades ending in .0 through .4 will be rounded down. Fractional portions of grades ending in .5 through .9 will be rounded up.

- No further adjustments will be made to grades. This policy applies even in situations where increasing a grade by just 1 point would cause a student's final letter grade for the course to cross a threshold (i.e. from B+ to A-). Regardless of the potential consequences, grade adjustments will be limited to automatic rounding.

Grading Breakdown

Category	Weight
Assignments	40%
In-class Quizzes	25%
Course Project	35%

Assignments: The three assignments will include both code and analyses to be uploaded for grading.

- To encourage students to plan ahead and manage their workload effectively, assignments submitted early will receive 1% extra credit per day, up to a maximum of 5%, with the final score capped at 100.
    - For example, if a student earns a raw score of 92 and submits the assignment 2 days early, their final score would be 93.8 (92 + 2%), and if the same student submits the assignment 7 days early, the bonus is capped at 5%, so the final score would be 96.6 (92 + 5%). In all cases, the final recorded grade will not exceed 100; for example, if a student earns a raw score of 98 and submits the assignment early, any bonus that would raise the score above 100 will be capped, and the final grade will be recorded as 100.
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- Late assignments will be accepted up to three days after the due date with a penalty of 10% per day.
    - For example, if the same student earns 92 but submits the assignment 1 day late, the final score would be 82.8 (92 − 10%). Assignments submitted more than three days late will not be accepted unless accompanied by a valid, officially documented excuse supported by the university.

In-class Quiz: There will be four short quizzes, each lasting at around 30 minutes during class time. The lowest quiz score will be dropped.

- For example, if your quiz scores are 60, 85, 95, and 90, the score of 60 will be dropped, and the average of 85, 95, and 90 will be used, resulting in an average quiz score of 90.

Course Project: The project begins with a short proposal outlining the research question, dataset(s), and the intended generative AI method, clearly tied to class topics (e.g., prompt engineering, fine-tuning, RAG, or advanced frameworks). Students will then carry out the plan, refining methods with feedback. A midterm check-in presentation will provide an opportunity for each team to share progress, receive feedback, and adjust their approach. Graduate students are expected to go further by integrating state-of-the-art methods and producing a more substantial deliverable. At the end of the semester, each team will present their work and submit

a final report summarizing the research question, dataset, approach, results, and reflections. Detailed guidelines and milestones will be provided on Canvas**.** No late submission is permitted for course project–related deadlines.

Course Schedule

This schedule is tentative and may be subject to change. You are responsible for staying updated on any changes communicated via Canvas.

Week	Dates	Topics	Quiz	Assignment	Project Milestone
1 (In-person)	Jan 26 & Jan 28	Introduction to Generative AI & LLMs	–	–	–
2 (In-person)	Feb 2 & Feb 4	Pretraining, Tokenization & Embeddings	–	–	–
3 (Zoom)	Feb 9 & Feb 11	Transformer Architecture 1	–	Assignment 1 Released (Transformer)	–
4 (Zoom)	Feb 16 & Feb 18	Transformer Architecture 2	Quiz 1	–	Project Requirement Released
5 (Zoom)	Feb 23 & Feb 25	Transformer Architecture 3	–	–	Project Idea Proposal Due
6 (Zoom)	Mar 2 & Mar 4	Prompting Engineering	–	Assignment 2 Released (Chain-of-thought)	–
7 (Zoom)	Mar 9 & Mar 11	Chain-of-Thought	Quiz 2	–	–
8 (Zoom)	Mar 16 & Mar 18	Evaluating GenAI/LLMs	–	–	–
9 (Zoom)	Mar 23 & Mar 25	*No class (Spring Break)*	–	–	–
10 (Zoom)	Mar 30 & Apr 1	Retrial Augment Generation (RAG) 1	Quiz 3	Assignment 3 Released (RAG)	Mid-Project Check-in Reports
11 (Zoom)	Apr 6 & Apr 8	Retrial Augment Generation (RAG) 2	–	–	–

12 (Zoom)	Apr 13 & Apr 15	Fine-tuning, Alignment, and Reasoning 1	–	–	–
13 (Zoom)	Apr 20 & Apr 22	Fine-tuning, Alignment, and Reasoning 2	Quiz 4	Assignment 4 Released (Fine-Tuning and Alignment)	–
14 (Zoom)	Apr 27 & Apr 29	Fine-tuning, Alignment, and Reasoning 3	–	–	–
15 (In-person)	May 4 & May 6	Final Project Presentations	–	–	–
16 (No meeting)	Finals Week	No Final Exam	–	–	Final Demo & Report Due

Additional Policies

Attendance Policy

Attendance is required but not graded. You are responsible for all material covered in class, regardless of attendance. Absences do not excuse missed assignments or announcements. If you have a documented excuse (e.g., athletics, SDS, or the Dean of Students), you will be exempt from penalties for missed graded activities as appropriate.

Participation

Active participation is expected in discussions and labs. Graduate students are expected to lead one advanced discussion topic.

Resources and Support for Students

As an instructor, I have a duty to report. This means I am required to promptly report to the Deputy Title IX Coordinator (titleix@miamioh.edu) any information a student shares with me regarding harassment, discrimination, sexual misconduct and interpersonal violence, or retaliation. A report does not initiate an investigation. It engages a discussion of your resources, supportive measures, and options available. If students want to speak with someone confidentially, the following resources are available on and off campus:

- Student Health Services, (513) 529-3000
- Student Counseling Services, (513) 529-4634
- Women Helping Women (WHW) Sexual and Interpersonal Violence Support Specialists are available to support all students and can be contacted by emailing

mu@womenhelpingwomen.org. As well as calling/texting 513-846-8402 between 9AM-5PM. The 24-hour hotline is 513-381-5610. WHW supports ALL survivors of dating/domestic violence, sexual assault, and stalking, regardless of gender identity or sexual orientation.

Speaking with a confidential resource person does not preclude students from making a formal report to the University if and when they are ready.

https://miamioh.edu/diversity-inclusion/programs-resources/report-incident/index.html

For more information, please visit https://miamioh.edu/campus-safety/sexual-assault/

and https://www.miamioh.edu/diversity-inclusion/oeeo/index.html

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Academic Integrity

All **assignments** must reflect your own understanding. Collaboration is permitted only when explicitly specified in the assignment instructions and must be clearly acknowledged. Any form of academic dishonesty, including the use of unauthorized AI tools, will be reported and may result in a failing grade or disciplinary action. Miami University's Academic Integrity Policy can be found here.

Note: For the **course project** only, students may use and adapt code written by others (including AI tools such as ChatGPT, or open-source repositories from platforms like Hugging Face or GitHub) as the foundation for their project, provided they follow these guidelines:

- Use of other people's material must not infringe the copyright of the original author, nor violate the terms of any licensing agreement. Know and respect the principles of fair use with respect to copyrighted material.
- Students must scrupulously attribute the original source and author of whatever material has been adapted for the assignment. Summarize the changes or adaptations that have been made. Make plain how much of the assignment represents original work.

Final Note

This syllabus provides a comprehensive overview of the course structure, expectations, and policies. Adjustments may be made throughout the semester, and all updates will be communicated via Canvas.